

Using the standard DEB animal model for toxicokinetic-toxicodynamic analysis

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ABSTRACT

Toxicokinetic-toxicodynamic (TKTD) models are essential tools to close the gap between laboratory toxicity tests and impacts on individuals under field conditions. For sub-lethal effects on life-history traits, dynamic energy budget (DEB) theory provides a solid basis for TKTD modelling. Various DEB-based TKTD models have been presented, ranging from complex and comprehensive to simple and reduced. Here we present a complete and up-to-date model formulation at the complex end of the range. The basic part of the model, governing the life cycle of the species without toxicant stress, is formed by the standard DEB animal model. This part can directly use the species-specific model parameters from the add-my-pet (AmP) library. Additionally, we cover one extension of the standard model with metabolic acceleration for some time after birth (the abj-model). The basic DEB model is augmented by a TKTD model that explicitly considers damage as driving force for the toxic effects, and which considers various feedbacks from life-history traits to damage dynamics (e.g., dilution by growth). Several choices are needed in this linkage, which are made explicit. A Matlab-based toolbox is presented to support the model analyses, which is illustrated with a case study for the springtail *Folsomia candida* exposed to chlorpyrifos in food.

1. Introduction

Laboratory toxicity testing is a mainstay of ecotoxicological effects assessment, both for scientific and regulatory purposes. However, the purpose of such experimental tests is not to say something about the laboratory situation, but rather to infer impacts on (populations of) individuals under field conditions. The classical dose–response analysis is unable to close the gap between the lab and the field because it cannot consider the factor ‘time’ nor deal with extrapolation across exposure patterns or environmental conditions (Jager, 2011). Closing the gap requires toxicokinetic-toxicodynamic (TKTD) modelling (Ashauer and Escher, 2010; Jager et al., 2006) that explicitly considers time as a factor, next to the exposure scenario, and builds on mechanistic principles. A model calibrated on a laboratory data set thereby offers the possibility to provide meaningful extrapolations to field conditions.

This potential was recognised by the regulatory community: in 2018, the European Food Safety Authority released a scientific opinion on the potential application of TKTD models for risk assessment of pesticides for aquatic organisms (EFSA, 2018). For sub-lethal effects on animals, models based on dynamic energy budget (DEB) theory

(Jusup et al., 2017; Kooijman, 2010) are the tools of choice. Routine application of DEB-TKTD models is, however, hampered by the fact that there is no single, unique model. Instead, there is a family of models, all closely related, ranging from complex and comprehensive to simple and reduced (Sherborne et al., 2020). The simplest DEB-TKTD model is based on a reduced theoretical framework (DEBkiss, Jager et al., 2013); a complete and up-to-date presentation of this model version was presented by Jager (2020). This model is the last incarnation in a long line of simplified, so-called DEBtox, models (e.g., Billoir et al., 2008; Jager and Zimmer, 2012; Kooijman and Bedaux, 1996).

At the other end of the scale, we can find the standard DEB animal model (Sousa et al., 2010). This model has mainly found application outside of ecotoxicology, such as in marine aquaculture and fisheries (Alunno-Bruscia et al., 2009), and forms the basis of the add-my-pet (AmP) library of parameter values (Marques et al., 2018). This library contains a complete set of parameter values for a wide range of species (at the time of writing, over 3700), including many of the popular test species used for environmental risk assessment. The standard model has had several applications in ecotoxicology (notably, Jager et al., 2010;

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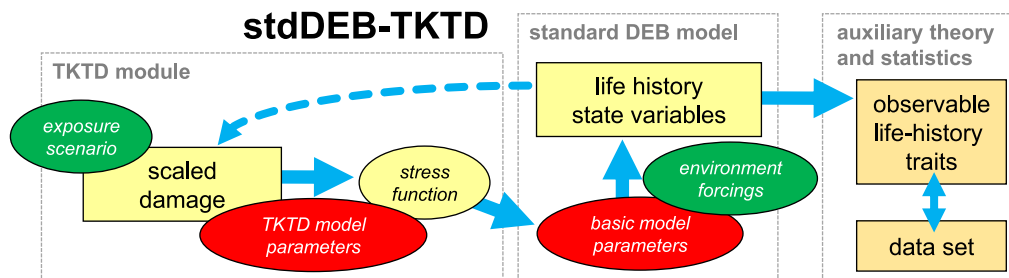


Fig. 1. Structure of the stdDEB-TKTD model. Solid arrows show the causal information flow in the model. The broken arrow represents potential feedbacks such as dilution of damage by growth. Red ellipses show model parameters, whereas green ellipses indicate scenario choices. Yellow boxes and ellipse represent the generic model structure, which (in principle) does not depend on the species or toxicant.

Jager and Selck, 2011; Augustine et al., 2012; Zimmer et al., 2018). The TKTD module was, in these cases, based on the earlier DEBtox models (Kooijman and Bedaux, 1996), focussing on an internal concentration of the compound, as driver of toxic effects. This module is outdated, and requires revision to bring it in line with the developments in survival modelling (see Jager, 2020), recognising the importance of considering ‘damage’ as the driving force of toxicity. Furthermore, the earlier work on standard DEB suffered from a lack of detail regarding the actual model equations, how they were implemented, and the underlying assumptions.

The purpose of this work is first to provide a complete set of equations for the standard DEB animal model, using the same parameters as provided in the AmP library, and including the links between model variables and measurable properties (auxiliary theory). We will include one model extension, namely metabolic acceleration from birth to ‘metamorphosis’ (the abj-model, see Marques et al., 2018). Secondly, we will link the standard DEB model to the updated TKTD module of Jager (2020). This requires a number of choices, which we will make explicitly. We will refer to this model version as stdDEB-TKTD. Thirdly, we present a Matlab-based toolbox for model calibration and model predictions, which will be illustrated with a case study. Finally, we discuss the open questions of the resulting model as well as the challenges in parametrising it.

2. Model description

2.1. Structure of the stdDEB-TKTD model

The general structure of stdDEB-TKTD is shown in Fig. 1. The model is build up from three modules. The standard DEB animal model holds a central position, flanked by a TKTD module and a module for auxiliary hypotheses and statistics. This latter module deals with the translation of the (abstract) state variables of the model to measurable properties, and the statistical model needed to fit the model to observations. The TKTD module is a so-called reduced model (see Jager, 2020) in which toxicokinetics and damage dynamics are represented by a single compartment (scaled damage).

The full set of model equations is rather extensive, and its implementation also requires some explanation. For example, the initial values for the state variables and the costs for an egg are implicit, and need to be found by simulation. Therefore, we present the modelling details and the underlying assumptions in the supporting information. In the main text, we focus on the structure of the model and the case study. This is warranted as the equations of stdDEB-TKTD are not, by themselves, new, although they have never been published in this combination and with this level of completeness.

2.2. Standard DEB model

The energy flows in the standard DEB animal are schematically shown in Fig. 2. This model has been extensively discussed elsewhere

(e.g., Kooijman, 2010; Jusup et al., 2017; Sousa et al., 2010), so we keep the description to a minimum here, and focus on the key aspects for TKTD modelling and the open questions that need to be addressed.

The model is phrased in terms of powers (energy fluxes, J/d) and volumetric body length (cubic root of structural volume, cm), so that it can directly use the energy-based parameters from the AmP library (Marques et al., 2018). This model represents an animal that does not change in shape, with a single reserve compartment. Additionally, we implemented one extension of the standard model, namely the abj-model for metabolic acceleration (Marques et al., 2018), as an option. Many species accelerate their growth in early life, and this extension provides an explanation, accelerating metabolism between birth and ‘metamorphosis’, which occurs before or at puberty (the start of investment into reproduction). Please note that we still refer to this model extension as a standard DEB model, to contrast it with the simplified DEBtox model versions. Within the AmP context, the abj-model would be seen as an alternative to the standard model (as, in that context, simplified models are not considered).

In the current formulation, a reproduction buffer is not considered. Instead, the allocated flux for reproduction is instantly converted into eggs. The reasons to exclude the buffer are pragmatic: the AmP entries currently do not consider the reproduction buffer, and using these parameters in a model with buffer would lead to bias (see Jager et al., 2022). Furthermore, dealing with the build-up of the buffer and the discontinuities in the model at spawning events poses a range of problems for model implementation as well as for the interpretation of data, particularly when following a cohort of animals in a toxicity test (see Jager et al., 2022). Nevertheless, good examples of DEB models with explicit consideration of the reproduction buffer are available, specifically for vertebrates (e.g., Augustine et al., 2012; Marn et al., 2022).

Table 1 gives the basic model parameters: those that do not depend on the toxicant. This list contains a few non-standard parameters that need to be explained. Firstly, the maturity efficiency κ_H does not appear in standard DEB formulations. This parameter is by default set to 1, so it has no effect. However, it is needed to provide a handle for potential toxicant stress on the maturation process (Koch and De Schampelaere, 2021). The additional parameter for the starvation module (γ_p) is explained in the following section.

2.3. Starvation strategy

It is important for any TKTD model for sub-lethal effects to consider (periodic) situations of starvation. This is obvious when simulating situations with time-varying food availability, but toxicant stress can also lead to starvation, especially under pulsed exposure (Jager, 2020). Starvation is the situation where the somatic maintenance needs cannot be met with the standard set of rules in the model. This situation can occur, even under abundant food, with chemicals that affect the energy fluxes for assimilation or maintenance. In contrast to the simplified model (Jager, 2020), the standard DEB animal has a reserve to buffer

Table 1

Explanation of parameter symbols for the basic model (excluding toxicant stress). Default values for *Folsomia candida* from AmP (Hamda et al., 2016), except for y_p and κ_H . Several parameters were re-fitted to the control treatment of the case study data set. Confidence intervals (CI, 95%) calculated by profiling the likelihood function.

Sym.	Explanation	Unit	AmP value	Re-fitted (CI)
Primary parameters				
T_A	Arrhenius temperature	K	8000	Fixed
$\{ \dot{P}_{Am} \}$	Max. area-specific assimilation	J/(cm ² d)	84.56	Fixed
$\{ \dot{P}_M \}$	Volume-specific somatic maint.	J/(cm ³ d)	845.6	2030 (1620–2540)
$\{ \dot{P}_T \}$	Surface-specific somatic maint.	J/(cm ² d)	0	Fixed
$\dot{k}_J$	Maturity maint. rate constant	1/d	0.002	Fixed
$[E_G]$	Volume-specific costs for growth	J/cm ³	4314	Fixed
E_H^b	Maturity level at birth	10 ⁻³ J	5.193	3.15 (2.52–3.83)
E_H^J	Maturity level at metamorphosis	10 ⁻³ J	54.83	75.2 (53.7–96.9)
E_H^P	Maturity level at puberty	10 ⁻³ J	111.2	295 (238–390)
$\dot{v}$	Energy conductance	10 ⁻³ cm/d	3.881	6.61 (5.80–7.81)
κ	Allocation fraction to soma	–	0.2226	0.444 (0.396–0.504)
κ_R	Reproduction efficiency	–	0.95	Fixed
κ_H	Maturation efficiency	–	1	Fixed
Parameters for provisional starvation module				
y_P	Yield factor for shrinking	J/J	0.64	Fixed
Forcings, environmental conditions				
f	Scaled functional response	–	0.9	Fixed
T_{ref}	Reference temperature (20 °C)	K	293.15	Fixed
T	Actual temperature (20 °C)	K	293.15	Fixed
Conversions				
d_V	Dry-weight density of structure	g/cm ³	0.17	Fixed
δ_M	Shape correction coefficient	–	0.1766	Fixed

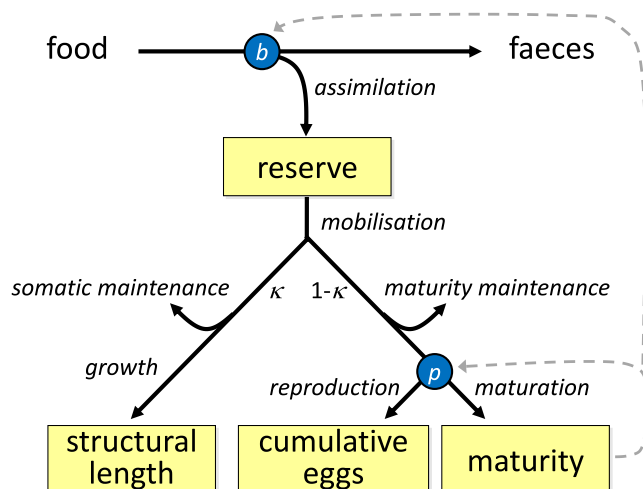


Fig. 2. Schematic representation of the standard DEB animal model. The nodes b and p denote switches at birth and puberty, governed by maturity thresholds (broken arrows).

the resource-requiring processes from rapid changes in food availability (and toxicant effects on assimilation and maintenance). However, it is good to note that this buffering can only help animals that are not yet fully grown. For a fully-grown adult, even a tiny decrease in assimilation, or a tiny increase in somatic maintenance costs, would trigger a starvation response (see also [Van der Meer, 2016](#)).

Under starvation, the animal needs to deviate from the standard rules in the model. The exact strategy that is followed is species specific (Kooijman, 2010, Page 118). For the current formulation, we selected the same strategy as used in the simplified model (Jager, 2020). When somatic maintenance needs cannot be met from the mobilisation flux allocated to the soma (the κ flux in Fig. 2), growth stops and the required resources for somatic maintenance are recovered from the $1-\kappa$ flux. Maturity maintenance is paid as long as possible, so reproduction may stop before the entire $1-\kappa$ flux is needed for somatic maintenance. When the $1-\kappa$ flux decreases even further, maturity maintenance is not being (fully) covered anymore. This is in line with the suggestion

of Kooijman (2010, Page 50) that maturity maintenance is more facultative than somatic maintenance. We assume that, within the context of ecotoxicological questions, this reduction in maturity maintenance has no negative consequences for the individual. When even the entire mobilisation flux is insufficient to pay somatic maintenance costs, the animal will pay these costs from structure, and thereby shrink. An additional yield parameter y_p is needed for shrinking under starvation. The transformation of reserve into structure comes with a certain loss of energy, which is included into the growth costs $[E_G]$. However, these overheads are not recovered when structure is used for maintenance. In fact, we need to account for additional overhead costs. The value of y_p in Table 1 assumes that both transformations have an 80% yield (following Jager, 2020).

This set of starvation rules has the benefit that it allows somatic maintenance to be paid in full under all circumstances, and that it leaves the reserve dynamics intact. Given the proposed surface-limited mechanism for reserve mobilisation (Maino et al., 2014), modifying reserve dynamics under starvation seemed less plausible to us. This set of rules has the, possibly unrealistic, property that an animal will shrink while still possessing a substantial reserve compartment. Different starvation strategies can be put forward, but it is good to realise that the details of the starvation module are unlikely to become important in a TKTD context: starvation will usually be limited to pulse-exposure conditions where it will be intermittent only.

2.4. TKTD module

For stdDEB-TKTD, the standard DEB model is linked to the TKTD module as presented in Jager (2020). This module explicitly considers that effects are driven by damage rather than an internal concentration. Furthermore, it provides a series of optional feedbacks (broken arrow in Fig. 1). These feedbacks represent the influence of (changes in) life-history traits on toxicokinetics or damage dynamics. Three types of feedbacks are considered: dilution by growth, modification of rate constant due to changes in surface:volume ratio with growth, and losses with reproduction. Which feedbacks are relevant will depend on the compound and on the species.

Here, we only consider the so-called reduced TKTD module, where toxicokinetics and damage dynamics are lumped into a single compartment (see Fig. 1). In a full TKTD module, separate compartments would

be used, which generally requires measurements of body residues for a successful calibration. The TK part of such a full model may range from a simple one-compartment model to full-fledged physiologically-based models (e.g., Mavroudis et al., 2018). It is important to stress that the single damage compartment of the reduced module does not necessarily represent an internal concentration of the toxicant. Depending on the relative speeds of toxicokinetics and damage dynamics, it may represent an internal concentration (of the parent compound or a transformation product) or a form of damage. The feedback process of growth dilution is always relevant for internal concentrations, but not necessarily for damage. Therefore, these feedbacks are optional, and various configurations make sense (Jager, 2020).

Sub-lethal and lethal effects are considered, and by default both are assumed to be driven by the same damage level. For sub-lethal effects, damage affects one or more primary parameters of the energy budget model with a linear-with-threshold relationship. For lethal effects, damage affects the hazard rate, also with a linear-with-threshold relationship, but with different parameters from the sub-lethal effects. Following Jager (2020), we thus assume the stochastic-death interpretation for mortality, and ignore the alternative of individual tolerance (see supporting information). Mortality in the control is covered by a constant background hazard rate h_b . A constant hazard rate is often a reasonable assumption since many standard toxicity tests are too short to see age-related mortality. However, there are exceptions (e.g., multi-generation studies) where addition of a dedicated ageing module may be considered. In such cases, care is needed as toxicant effects may interfere with senescence (e.g., Jager et al., 2007).

The TKTD module was originally developed for the simplified model (Jager, 2020). In that model version, all body mass is treated as structure. In contrast, in stdDEB-TKTD, body mass is made up of structure and reserve. Since structure and reserve have their own dynamics, this requires some explicit choices regarding the feedbacks to damage dynamics. Clearly, this is an aspect where more detailed experimental work is needed, complicated by the uncertainty regarding the nature of damage, as outlined above. For now, we propose to use structural size for all feedbacks relying on body surface area and volume, and thus ignore the potential role of reserve. In other words: the module from Jager (2020) is used as is, using structural length measures for the feedback processes. The resulting module is simple to use, but its mechanistic interpretation in the presence of a substantial reserve compartment is rather awkward and can easily lead to inconsistencies (see supporting information, Section 2.1). Fully mechanistic alternatives will involve more parameters, and thereby require more information for model calibration. For example, it is possible to use a TK module that explicitly partitions the toxicant over structure and reserve (e.g., Augustine et al., 2012). Supporting such an extension would require additional information such as measured internal concentrations over time. However, if we cannot be sure that an internal concentration is driving the toxic effects, we also need to consider damage dynamics, and how damage partitions over structure and reserve. For now, we therefore propose to keep this module simple.

2.5. Physiological modes of action

In DEB-TKTD models, the physiological mode of action (pMoA) specifies the basic parameter(s) of the energy budget that are changed due to the damage caused by the toxicant. For the simplified model (Jager, 2020), a selection was made, partly based on consistency with previous work (Billoir et al., 2008; Jager and Zimmer, 2012; Kooijman and Bedaux, 1996), and partly based on limitations owing to the simplifications of the model. For example, the allocation fraction κ is hidden in simplified models (combined with other parameters into compound parameters). Therefore, there is no pMoA for changes in this allocation fraction in these models. Additionally, in simplified models, the pMoA maintenance increases both somatic and maturity maintenance to the same extent. This was done to guard consistency with DEB models that

Table 2

Basic physiological modes of action (pMoA) and how the stress factor s affects the primary parameters of standard DEB. The stress factor is a linear-with-threshold function of damage. Vector S is used to specify the pMoA, and also allows simple combinations.

pMoA	S	Change in target parameter(s)
Assimilation or feeding	[1,0,0,0,0]	$\{\dot{p}_{Am}\} \rightarrow (1-s)\{\dot{p}_{Am}\}$
Maintenance costs	[0,1,0,0,0]	$[\dot{p}_M] \rightarrow (1+s)[\dot{p}_M], k_J \rightarrow (1+s)k_J$
Growth/maturation costs	[0,0,1,0,0]	$[E_G] \rightarrow (1+s)[E_G], \kappa_H \rightarrow \kappa_H/(1+s)$
Reproduction costs	[0,0,0,1,0]	$\kappa_R \rightarrow \kappa_R/(1+s)$
Hazards to embryo	[0,0,0,0,1]	$\kappa_R \rightarrow \exp(-s)\kappa_R$

explicitly include a maturity state variable. Simplified models use body length rather than maturity to switch between life stages, and when somatic and maturity maintenance are changed by the same factor, body length remains a good proxy for maturity status.

In stdDEB-TKTD, we do not have to worry about these limitations. We can assume that the toxicant affects κ or only somatic maintenance (see Jager et al., 2010), and watch length at puberty shift under stress. The stdDEB-TKTD model is phrased in primary parameters, which makes it easy to apply stress to fundamental processes in the energy budget: the energy flows in Fig. 2. We can thus dream up a long list of potential pMoAs, as, in principle, any DEB parameter may be modified by a stressor, and even combinations can be plausible (i.e., a toxicant affecting more than one parameter). However, we propose to start from the classical limited set, analogous to the pMoAs used in the various simplified models (Table 2). To ensure that the pMoA costs for growth does not produce large shifts in length at puberty, we need to also apply the stress factor to the costs for maturation. This was not necessary in simplified models as they do not follow maturity as a state variable. In principle, all pMoAs in Table 2 lead to a rather constant length at puberty. However, exact constancy is only guaranteed for a specific choice of the model parameters governing somatic and maturity maintenance.

2.6. Auxiliary theory and statistics

The state variables of the basic DEB model (i.e., excluding the TKTD module) are reserve (J), volumetric structural length (cm), maturity level (J), and cumulative egg production (eggs). Body size in experimental toxicity tests is commonly determined as body length. In those cases, we can use the shape correction coefficient δ_M to translate volumetric length of the model into actual length. The use of a constant δ_M is warranted when the organism does not change in shape over ontogeny (at least that part of ontogeny that is relevant to the model analysis), and when the actual length measure is not affected by the individual's reserve density. Determining body size as volume or wet weight is generally more robust, though the calculations are slightly more complicated since these metrics have a contribution from both structure and reserve. The energy in the reserve thus has to be translated into a volume or weight, and added to the contribution from structure. This translation is presented in detail in the supporting information. The cumulative egg production in the model can be directly compared to observed cumulated eggs, under the assumption that the transformation from allocated reserve to egg is instantaneous and continuous. Special care is needed for species that spawn clutches of eggs at considerable time intervals, and for species that incubate the eggs in a brood pouch (Jager et al., 2022).

For the statistical treatment, we can follow the likelihood principles laid out in Jager and Zimmer (2012) (and see supporting information). For survival data, the multinomial distribution is used, while for sub-lethal data, the residuals are assumed to follow independent normal distributions, if needed after transformation. In general, it is advisable to apply transformations for sub-lethal endpoints, to bring the error structure more in line with a normal distribution (i.e., correcting residual variance increasing with the mean). However, it is

good to realise that independence is usually violated for growth and reproduction data. The log-likelihood contributions from different traits (i.e., survival, growth and reproduction) are added, again assuming independence. The overall likelihood function can subsequently be used in optimisation and uncertainty analysis in a Bayesian or frequentist framework.

2.7. Implementation in code

The stdDEB-TKTD model was implemented into the BYOM platform under Matlab (<http://www.debtox.info/byom.html>). BYOM is a flexible framework for simulating and fitting models, and includes various methods for optimisation, uncertainty analysis, and uncertainty propagation to model predictions. The integration of the stdDEB-TKTD model into BYOM implies that all of the code developed for other types of models can be used, including advanced statistical treatment (Jager, 2021) and moving time window extrapolations for complex exposure scenarios (Sherborne et al., 2020, 2022). This code is freely available from the BYOM web page, and the stdDEB-TKTD package (version 1.0) includes the code to re-run the case study, presented in Section 4.

3. Open questions

Turning the standard DEB animal model into a TKTD model leads to a number of questions that still remain to be addressed. In Section 2.4, we already pointed at limitations in the TKTD module due to ignoring the role of reserve in toxicokinetics and damage dynamics. In Section 2.5, we discussed that stdDEB-TKTD allows for a much wider range of potential pMoAs than the simplified DEBtox models. In this section, two additional issues are discussed.

3.1. Maternal effects and initial states

The maternal effects rule in DEB theory states that a mother tunes the cost of a single egg in such a way that it leads to an offspring that, at birth, has the same reserve density that its mother had at egg formation. This rule implies that egg costs depend on reserve density of the mother, and thus on food availability. Mothers in poor conditions produce eggs with less reserve, that yield offspring in poor condition. This rule dictates the egg costs (and thereby affects the reproduction rate in eggs/d) but also the initial values for the state variables. Regarding the latter, we would need to make assumptions about the feeding status of the mothers that produced the eggs that spawned the individuals that were used to start the toxicity test. Unfortunately, there is no explicit relationship between the egg costs and the mother's reserve density, so application of this rule requires implementing some search algorithm.

This rule, however, requires further discussion. Firstly, for some species, it is at odds with biological observations. For example, for *Daphnia magna*, poorly-fed mothers actually produce larger offspring, and offspring size also increases over consecutive broods (Gabsi et al., 2014). Secondly, applying the maternal effects rule strictly implies that we continuously need to recalculate the egg costs over the course of the experimental test, as reserve density of the mother may change over time, specifically when the toxicant affects assimilation or when the toxicant stress induces starvation. Furthermore, we need to consider whether the embryo is affected by the toxicant stress as well, and whether the mother then compensates for that stress by adapting the egg costs to maintain the maternal effects rule. To test the maternal effects rule in ecotoxicological practice would require following the energy content or size of eggs or offspring in toxicity tests. At least for invertebrate test species, this is not common practice.

At this moment, the BYOM code includes various options to deal with the maternal effects rule, so that the user can try different interpretations. There are three options for egg costs: calculate egg costs from food level f in the toxicity test, or always from $f = 1$, or using the

actual reserve density e of the mother. Toxicant stress on the embryo is not considered for the egg-cost calculation. For the initial states at the start of the toxicity experiment, there are similarly two options: calculate the states from the food level f in the toxicity test, or always from $f = 1$. Which of these options is most realistic will depend on the specific culturing and testing conditions for the species, in each case. In general, we should not expect model results to be substantially affected by this choice: as long as f is not extremely low, and the test starts with neonates, we should rapidly see a steady state for the reserve density with the food level in the test. Nevertheless, this is a choice that needs to be made explicitly. For both egg costs and initial states, we suggest using f for the toxicity test as default. Often, we will start the model at birth, but the code also offers the possibility to start at a specified body size after birth. This is needed for toxicity tests that do not start with early neonates.

3.2. Use of AmP entries

Contrary to simplified models, the standard DEB model cannot be completely parametrised from the controls of a toxicity test; information on growth and reproduction is simply insufficient to identify all model parameters. For this reason, the add-my-pet (AmP) library (Marques et al., 2018) has been proposed to support TKTD modelling with standard DEB (Baas et al., 2018; Zimmer et al., 2018). The quality and quantity of the data underlying the entries varies, which is reflected in several marks provided with each entry. For the standard test species in aquatic risk assessment, the entries are generally of high quality. Entries can be updated at any time to reflect new knowledge, subject to evaluation by the curators, and version control is in place.

Despite the obvious scientific value of this library, there is an important limitation in its application to TKTD analysis. This limitation results from the fact that each AmP entry is a compromise, attempting to reconcile various data sets from different sources. For TKTD analysis, in contrast, we need to represent the specific cohort of individuals in a specific toxicity test. Even for the highest-quality entries, it is safe to predict that the generic AmP parameter values for a species will not provide a precise correspondence to the control of a specific toxicity test. Given the variation in control response between experimental tests, even from the same lab, mismatch is almost guaranteed. However, a close correspondence to the controls is crucial for TKTD modelling, since toxic effects need to be identified as deviations from the modelled control situation.

Firstly, it is important to tune the environmental scenario in the DEB model (food availability via f , and temperature via T , see Table 1) to match the conditions in the toxicity test as closely as possible. For temperature, this should not cause too many problems since temperature can be measured unambiguously. For f , however, the situation is less clear, since this is a scaled parameter, scaled to some maximum achievable assimilation rate, and because f also depends on food quality. Therefore, it is not such a stretch of the imagination to assume that f in a specific toxicity test differs from the value used to construct the AmP entry, even under (presumed) *ad libitum* food conditions. The first step to address a mismatch would thus be to see if fitting only f to the control data of a toxicity test suffices to capture the observations. This strategy was followed for a fish early-life stage test by Zimmer et al. (2018). However, in that case, the data set only contained body-size measurement at the end of the test, which is insufficient to evaluate the adequacy of this strategy.

If a change in f is not sufficient to fit the control data, the AmP parameters would need to be modified to some extent. Since the control data by themselves are insufficient to refit all parameters, we either need to add more data (e.g., taken from the AmP entry), or re-fit only a selection of parameters. Adding more data is a troublesome procedure, also because data from other labs may not necessarily be consistent with the specific data set that we want to analyse. Fitting a selection of parameters may easily produce a good fit to growth and reproduction

in the control for the wrong reason. For example, yielding a parameter set implying unrealistic properties for unmeasured endpoints such as egg size, egg incubation time, body weight, ingestion rate, etcetera.

At this moment, there is no single strategy that we can advise for TKTD analysis. When modification of an AmP entry is needed to analyse results from a specific toxicity test, due care is needed, and the procedure would need to be properly documented. It is currently unclear to what extent TKTD model calibration and prediction would actually be affected by ‘wrong’ values for the basic parameters. It is conceivable that a good description for control growth and reproduction suffices for a proper calibration to toxicity test data, irrespective of the details of the underlying parameters. However, extrapolations to different exposure scenarios or environmental conditions may suffer more from an unrealistic basic parametrisation.

Finally, it is good to note that AmP entries do not yet account for a reproduction buffer and brood-pouch incubation. For some species, this can easily cause timing mismatches between the model and the observations, resulting in bias in the model parametrisation (see Jager et al., 2022).

4. Case study

To illustrate the model, its implementation, and potential application, we use a toxicity test with the springtail *Folsomia candida*, exposed to chlorpyrifos in food. This data set originates from Crommentuijn et al. (1997) and was analysed with a simplified model in full detail by Jager et al. (2007), and in a more limited form by Jager (2020). Here we use the reduced data set as used in Jager (2020), which implies that the data for reproduction and survival are truncated to the first 45 days of the test. These endpoints were affected by old age after longer durations, including interactions with the toxicant, which would require additional modules to deal with.

For this case study, we use frequentist, likelihood-based inference. For body weight and cumulative reproduction, normal distributions are used as approximation for the residuals, so we also need to consider applying transformations of model and data. For body weight data, we decided to use log-transformation, and for cumulative egg production square-root transformation. In Jager (2020), square-root transformation was used for both endpoints, but body size was expressed in that study on a length basis, as cubic root of wet weight. For body weight, a stronger transformation is thus appropriate. Regarding the maternal effects rule, we decided to calculate both egg costs and initial states from the food level as established for the toxicity test ($f = 0.9$), and to start the model at birth. The initial body weight of the animals at the start of the test (1-day old) is included as observation into the data set. Finally, control and solvent control are kept separate (shown with different symbols), but are both used for the control fits.

The model calibration is performed in three steps. First, several of the basic model parameters are fitted to the controls for growth and reproduction. This is done by simplex optimisation, followed by likelihood profiling. Next, the background hazard rate is fitted to the controls for survival. This is done separately from the sub-lethal controls, since background mortality is treated as an independent endpoint, independent from the energy budget. Finally, the control parameters (Table 1 and background hazard from Table 3) are kept fixed while fitting the toxicity-related parameters (Table 3) to the full data set. This latter optimisation is performed using the parameter-space algorithm laid out in Jager (2021), yielding optimised values, parameter confidence intervals, and a sample for uncertainty propagation.

4.1. Re-fitting AmP parameters

The AmP parametrisation for *F. candida* (Hamda et al., 2016) is given in Table 1. The AmP entry also included controls from Crommentuijn et al. (1997), and sets $f = 0.9$ for those data. We therefore use the same value in this study. The toxicity test was conducted at the

Table 3

Parameters for the TKTD and survival module, with parameter estimates for the case study (see model fit in Fig. 4). Confidence intervals (CI, 95%) calculated by profiling the likelihood function. Note that the background hazard h_b was fitted to the control treatments only, and kept fixed while fitting the five toxicological parameters.

Sym.	Explanation	Unit	Value (CI)
Toxicity parameters			
k_d	Dominant rate constant	1/d	0.179 (0.111–0.313)
S	Vector for pMoA	[–]	[0,0,0,1]
X	Vector for feedbacks	[–]	[0,0,0,0]
z_b	Threshold sub-lethal effects	mg/kg	5.86 (3.19–6.90)
b_b	Effect strength energy budget	kg/mg	0.237 (0.147–0.318)
z_s	Threshold lethal effects	mg/kg	6.45 (3.90–7.95)
b_s	Effect strength lethal effects	10^{-3} kg/mg/d	4.92 (2.91–7.48)
h_b	Background hazard rate	10^{-3} 1/d	2.65 (1.14–5.11)

Table 4

Four typical configurations for the damage module, with relative goodness-of-fit shown as difference in Akaike Information Criterion relative to the best fit (ΔAIC). Fits made with the pMoAs costs of reproduction and hazards to the embryo.

Scaled damage representation	X	ΔAIC costs	ΔAIC hazards
Damage that is not diluted	[0,0,0,0]	3.3	0
Full TK (incl. losses with repro.)	[1,1,1,1]	19	19
Damage that is diluted	[0,0,1,0]	19	15
Classical DEBtox TK	[1,1,1,0]	20	19

reference temperature, and therefore no temperature corrections were needed.

The AmP parameter set provides a very poor correspondence to the control for the case study (Fig. 3). Clearly, a change in f would be insufficient to obtain a good fit. The correspondence is so poor that identification of toxicant effects will be compromised. Therefore, we need to re-fit some of the basic model parameters to improve the fit. For this case study, we settled on re-fitting $[p_M]$, κ , E_H^b , E_H^j , E_H^p and $\dot{v}$ to the control response for growth and reproduction. This selection was inspired by the advice in Kooijman et al. (2008), and turns out to provide an excellent fit with all fitted parameters properly identifiable from the data (see Table 1 and likelihood profiles in supporting information). As explained in Section 3.2, the strategy for re-fitting AmP parameters requires further thought. However, for the case study, this selection suffices to demonstrate the functioning of the model.

The size of an egg is not a model parameter itself but rather a consequence of the parameters in Table 1. To avoid re-fitting to establish a parameter set implying biologically unrealistic egg sizes, we use the egg dry weight of 0.41 μg , as reported by Hamda (2014), as a constraint. The BYOM platform has a simple facility to include such constraints, similar to the use of zero-variate data in the AmP entries (Marques et al., 2018). It requires the definition of a normal distribution for the data point, here a mean egg weight of 0.41 μg with an arbitrary s.d. of 5% of the mean. The BYOM platform will apply this distribution as an additional, independent, likelihood function, combined with the other likelihood functions.

4.2. Fitting TKTD parameters

Firstly, the background hazard rate h_b is fitted to the control survival over time (Table 3). Next, the five toxicity-related parameters are fitted on the full data set (Table 3). For this case study, we focus on the pMoA's that provide a direct effect on reproduction, and on the four most obvious feedback configurations (Jager, 2020). The resulting goodness-of-fit is shown in Table 4. Results for more pMoA-feedback configurations are provided in the supporting information.

For this selection of pMoA-feedback configurations, the best fit is provided by ‘hazards to the embryo’, without any feedbacks. However, the fit is visually and statistically very similar to the fit with ‘costs for reproduction’, also without feedbacks. This latter configuration was used in Jager (2020), where the hazards pMoA was not considered. The

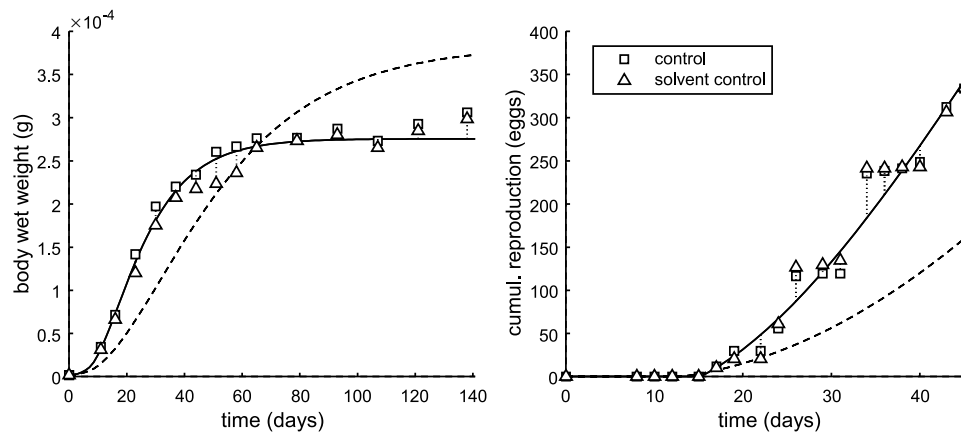


Fig. 3. Correspondence of the AmP parameter values to the controls of the case study (broken lines). Solid lines are model curves after re-fitting several parameters (see text) to provide a closer correspondence to the observations.

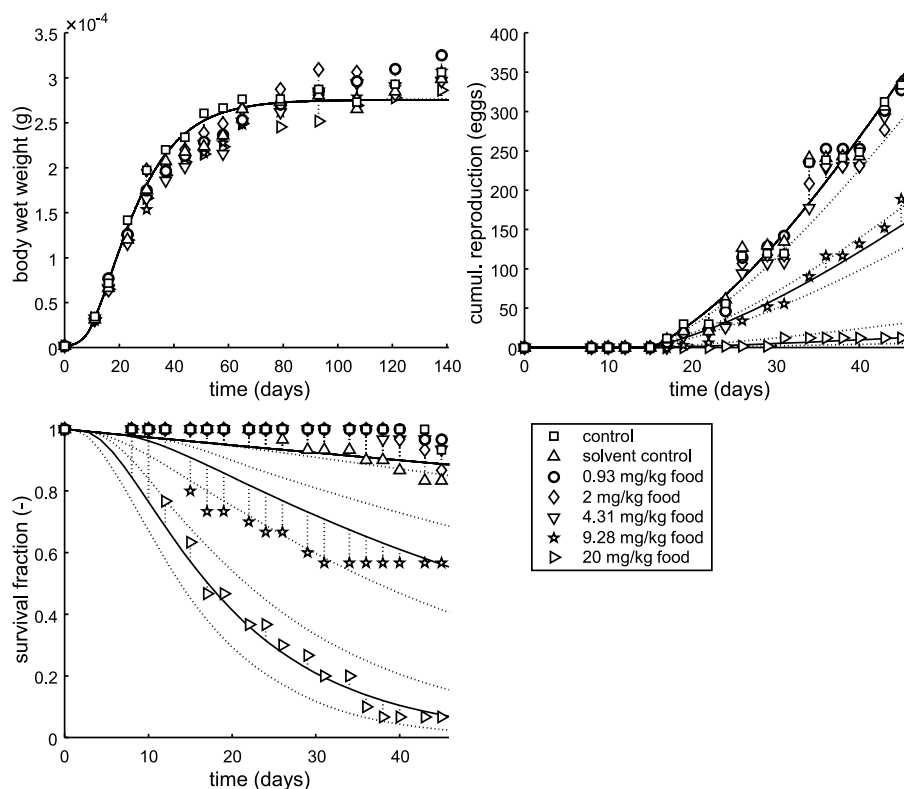


Fig. 4. Model fit to the full data set for the case study. Dotted vertical lines link symbols to the corresponding model curve. Dotted lines around the model curves show the 95% confidence interval resulting from uncertainty in the estimation of the fitted toxicity parameters. Parameter estimates provided in Table 3.

difference in AIC between the two pMoAs is very small and unlikely to be meaningful. For the best-fitting configuration, the overall fit to the data is very good (Fig. 4). The fit is almost identical to the one for the simplified model (Jager, 2020), though the visual comparison is hampered by the different expression for body size (wet weight versus volumetric length) and the fact that the simplified model was fitted without acceleration of metabolism.

More details on the model calibration are presented in the SI, including the parameter-space plot (see Jager, 2021) and sensitivity analyses. The optimisation yields a sample from parameter space that can subsequently be used to propagate uncertainty in the parameter estimates to model predictions such as model curves (see Fig. 4), EC_x values at different time points, or multiplication factors for exposure profiles derived from fate models (Sherborne et al., 2022).

5. Discussion

DEB-TKTD encompasses a family of closely-related models, ranging from simple to complex (Sherborne et al., 2020). A fully consistent and up-to-date model at the simple end of the range was presented recently (Jager, 2020); here, we present an up-to-date model formulation at the complex end of the range, based on the standard DEB animal model. Since these two model versions now use the same TKTD module, this facilitates comparison of their performance. Comparability is further facilitated by the fact that both model versions are now implemented into the same BYOM framework, and can thereby use the same numerical routines for calibration, prediction, and uncertainty propagation. At this moment, it is not clear which applications require stdDEB-TKTD and for which applications a simplified model version would suffice. In

general, selection of the most-suitable model depends on the purpose for which the model is to be used.

In the model formulation and implementation, we made a number of choices to apply the standard DEB animal model to TKTD analysis (see supporting information). This includes decisions on how to interpret the maternal effects rule, how to derive initial values for the state variables, how to include feedbacks on damage dynamics, and selecting a strategy for (toxicant-induced) starvation. Different choices are possible and defensible, but it is important to make these choices explicit. As the case study illustrates, application of stdDEB-TKTD also requires a number of explicit decisions to be made. The main issue here is the uncertainty of how to modify the AmP entry to fit the controls of the specific toxicity data set. Clearly, there is the risk of obtaining a good fit for the wrong reasons. Streamlining this step of the analysis would facilitate more routine use of the model.

Many questions in ecotoxicology, both scientific and regulatory, are best addressed by TKTD modelling. With our presentation of stdDEB-TKTD and its implementation, and by making the underlying design choices explicit, we hope to contribute to, and inspire, further development of this field.

CRedit authorship contribution statement

Tjalling Jager: Methodology, Software, Formal analysis, Writing – original draft. **Benoit Goussen:** Writing – review & editing, Project administration. **André Gergs:** Conceptualization, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Tjalling Jager reports financial support was provided by Bayer AG.

Data availability

Matlab code and data are published as part of the stdDEBtktd package on the BYOM web page (link provided in the manuscript).

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ecolmodel.2022.110187>.

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